

# Physics 4140H: Assignment I

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## Problem 1.

- (a) Find the curves on which the functional

$$\int_0^1 (y')^2 + 12xy dx$$

with  $y(0) = 0$  and  $y(1) = 1$  can be extremized.

- (b) On which curve can the function

$$\int_0^{\pi/2} (y')^2 - y^2 + 2xy dx$$

with  $y(0) = y(\pi/2) = 0$  be extremized?

- (c) Find the extrema of

$$\int_{x_1}^{x_2} (1 + x^2 y') y' dx.$$

## Solution 1.

- (a) Solving the Euler-Lagrange equation for  $f(x) = (y')^2 + 12xy$ ,

$$\frac{\partial f}{\partial y} = \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right).$$

Here we have

$$\frac{\partial f}{\partial y} = 12x; \quad \frac{\partial f}{\partial y'} = 2y'; \quad \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) = 2y''.$$

Hence we say that

$$6x = y''$$

and integrate twice to obtain

$$y = x^3 + Cx + D$$

and  $y(0) = D = 0$  and  $y(1) = 1 + C = 1 \implies C = 0$  so

$$y(x) = x^3$$

- (b) Again we apply the Euler-Lagrange equation with  $f = (y')^2 - y^2 + 2xy$  and obtain

$$\frac{\partial f}{\partial y} = 2x - 2y; \quad \frac{\partial f}{\partial y'} = 2y'; \quad \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) = 2y''.$$

So we say that

$$x = y'' + y.$$

First we solve the homogenous form  $y'' + y = 0$  by guessing a solution of the form  $e^{mx}$  which gives a characteristic equation  $m^2 + 1 = 0 \implies m = i$  and so we obtain  $y_h = Ce^{ix} = C \cos(x) + C_1 \sin(x)$ . Then we solve for the particular solution by guessing at the method of undetermined coefficients with  $y_p = x$  (which works). Hence the full solution is  $y = C \cos(x) + C_1 \sin(x) + x$ . Applying the boundary conditions,  $y(0) = C = 0$  and  $y(\pi/2) = C_2 + \pi/2 = 0 \implies C_2 = -\pi/2$  hence

$$y(x) = x - \frac{\pi}{2} \sin(x)$$

(c) Here

$$\frac{\partial f}{\partial y} = 0; \quad \frac{\partial f}{\partial y'} = 1 + 2x^2(y'); \quad \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) = 4x(y') + 2x^2(y'').$$

Hence

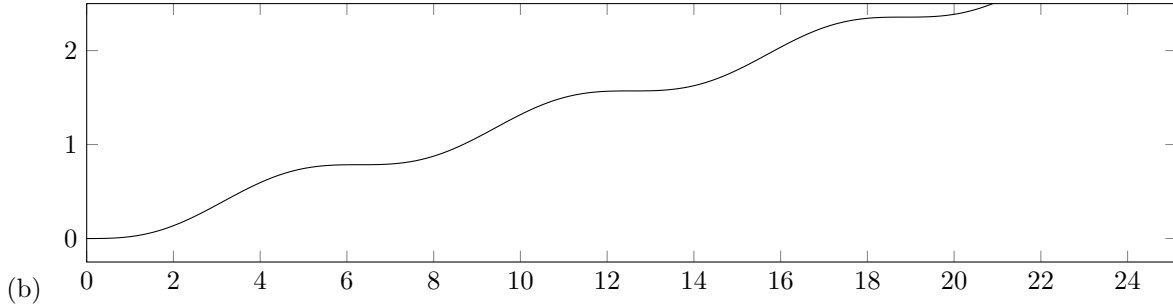
$$2xy' + x^2y'' = 0$$

which is a homogenous Cauchy-Euler equation and hence we take the solution to be of the form  $y = x^m$  and obtain the characteristic equation  $m^2 + 2m = 0 \implies m = 0, -2$ . Hence  $y = C_1 + C_2x^{-2}$ .

### Problem 2.

- (a) Review the brachistochrone problem (Example 6.2). Show that the time required for a particle to move (frictionless) to the minimum point of the cycloid is  $\pi\sqrt{a/g}$  where “a” is as shown in the figure 6-4 and “g” is acceleration due to gravity. (Hint: Use your knowledge of equations of a cycloid from Example 6.2; set up the problem using the time from kinematic equations; just before minimization convert  $x$  and  $y$  in terms of  $\theta$ , using the equations of a cycloid).
- (b) Plot the brachistochrone path in a Euclidian plane for  $\theta$  varying from 0 to  $8\pi$ .

**Solution 2.** (a) Review the brachistochrone problem (Example 6.2). Show that the time required for a particle to move (frictionless) to the minimum point of the cycloid is  $\pi\sqrt{a/g}$  where “a” is as shown in the figure 6-4 and “g” is acceleration due to gravity. (Hint: Use your knowledge of equations of a cycloid from Example 6.2; set up the problem using the time from kinematic equations; just before minimization convert  $x$  and  $y$  in terms of  $\theta$ , using the equations of a cycloid).



**Problem 3.** Fermat’s Principle: Consider light passing from one medium with index of refraction  $n_1$  into another medium with index of refraction  $n_2$ . Use Fermat’s principle and calculus of variation to minimize time and derive the Snell’s law of refraction:  $n_1 \sin \theta_1 = n_2 \sin \theta_2$ . Geometry based answers will receive zero points and are not accepted.

### Solution 3.

**Problem 4.** Find the shortest path between two points  $(x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$  in a 3-dimensional Euclidean space.

**Solution 4.** I assume you want a proof using calculus of variations but to be clear, distance in Euclidean space is defined as the length of the line segment, you can’t get smaller than that by definition.

### Problem 5.

- (a) Learn example 6.1. Plot the neighbouring function using the definition given in the book:  $\eta(x)$ ,  $y(x)$ , and  $y(\alpha, x)$  within the specified limit. Finally obtain  $J(\alpha)$  and argue that it must be higher than extremum value of  $J$ .
- (b) Compare a catenary path and a parabolic path by plotting them in the same graph.

### Solution 5.